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### DISCHARGE CAP

#### Abstract

The present invention has an object to suppress an external force applied to a container to discharge a content liquid CLq to an appropriate magnitude, to prevent the content liquid CLq discharged from a discharge nozzle **29** from becoming too thick, and to enable the content liquid CLq after discharge to be cut off in an improved manner. A valve part **49** included in a slit valve **7A** includes a slit part **49b** including slits **55**, and an outer circumferential part **49a** coupled to a thin-walled tube part **51** (a connecting part). The slit part **49b** more easily deforms with a pressure change in the interior space of the container than the outer circumferential part **49a**. The slit part **49b** deforms in a discharge direction of the content liquid CLq at the time of discharge of the content liquid CLq, and deforms toward a nozzle flow path **29a** after the content liquid CLq is discharged.

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## **Background/Summary**

### TECHNICAL FIELD

[0001] The present invention relates to a discharge cap.

### BACKGROUND

[0002] A viscous liquid, such as a lubricant, a caulking agent, or honey is stored in, for example, a container that has an elasticity and to which a discharge cap is attached at a mouth part. The discharge cap is provided with a nozzle for discharging a liquid (a content liquid) stored in the interior space of the container. The nozzle is a hollow tubular body where a flow path (a nozzle flow path) for a liquid is formed in the inner part thereof. The nozzle flow path is communicated with the interior space of the container when the discharge cap is attached to the mouth part of the container.

[0003] An external force is applied to the container when the content liquid is to be discharged from the nozzle. For example, when the container is compressed after the nozzle flow path is filled with the content liquid, the pressure in the interior space of the container is increased and the content liquid is discharged from the nozzle.

[0004] As such a discharge cap, for example, a discharge cap described in Patent Literature 1 is known. In the discharge cap described in Patent Literature 1, a nozzle includes a slit valve having slits in the shape of a cross formed thereon.

[0005] When the discharge cap in Patent Literature 1 is attached to a mouth part of a container, the nozzle flow path is filled with a content liquid, and an external force is then applied to the container, the content liquid in the nozzle flow path is pushed toward the slit valve with a pressure increase in the interior space of the container, and the slits in the shape of a cross are opened by the content liquid. Subsequently, the content liquid is discharged outside through a gap (a discharge port) of the slits.

[0006] When the external force to the container is canceled after the content liquid is discharged, the container inflates due to the elasticity of the container and the content liquid and air near the slits are drawn (sucked back) into the nozzle flow path. When the container returns to the original shape, the slits are closed.

[0007] In a container to which the discharge cap is attached, an external force to discharge the content liquid is required not to be too large. For example, even when the content liquid is a high-viscosity liquid such as a lubricant or honey, it is required that the content liquid is discharged with application of an appropriate magnitude of an external force to the container.

[0008] Elongating the slits provided in a valve to suppress the magnitude of an external force applied to the container is conceivable. However, simply elongating the slits may cause another problem that the content liquid discharged from the nozzle becomes too thick. Furthermore, a problem that the content liquid is not well cut off after discharge may also occur. For example, after the external force to the container is canceled, the content liquid that cannot be returned to the nozzle flow path may form a thread and adhere to the outer surface of the cap.

### CITATION LIST

#### Patent Literature

[0009] Patent Literature 1: Japanese Patent No. 6345040

## SUMMARY

### Technical Problem

[0010] As described above, the conventional discharge caps have a problem that a content liquid discharged from a nozzle becomes too thick when slits are elongated to suppress an external force applied to a container.

[0011] The present invention has been made in view of the problems described above, and a primary object thereof is to suppress an external force applied to a container to discharge a content liquid to an appropriate magnitude, to prevent the content liquid discharged from a nozzle from becoming too thick, and to enable the content liquid after discharge to be cut off in an improved manner.

### Solution to Problem

[0012] In order to solve the above problem, the present invention provides a discharge cap that is attached to a mouth part of a container having an elasticity and that discharges a content liquid associated with a pressure increase in an interior space of the container due to an external force applied to the container, the discharge cap comprising: a nozzle that forms a nozzle flow path with a sectional area smaller than an opening area of the mouth part; and a valve that is provided on the nozzle and that is made of an elastic material softer than the nozzle, wherein the valve includes a valve part through which the content liquid in the nozzle flow path is discharged, a tubular held part that is held on the nozzle by a holder, and a tubular connecting part that connects the valve part and the held part to each other, the valve part includes a slit part provided with a slit opening and closing according to a difference between an atmospheric pressure and a pressure in the interior space of the container, and an outer circumferential part coupled to the connecting part, and the slit part more easily deforms with a pressure change in the interior space than the outer circumferential part, deforms in a discharge direction of the content liquid at a time of discharge of the content liquid, and deforms toward the nozzle flow path after the content liquid is discharged.

### Advantageous Effects of Invention

[0013] According to the present invention, the external force applied to the container to discharge the content liquid can be suppressed to an appropriate magnitude, the content liquid discharged from the nozzle is prevented from becoming too thick, and the content liquid after discharge is enabled to be cut off in an improved manner.

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## Description

### BRIEF DESCRIPTION OF DRAWINGS

[0014] FIG. 1 is a sectional view illustrating a discharge cap and a part of a container according to a first embodiment, and illustrating a closed state where a lid body is placed thereon.

[0015] FIG. 2 is a sectional view illustrating the discharge cap and a part of the container according to the first embodiment, and illustrating an open state where the lid body is removed therefrom.

[0016] FIG. 3 is an enlarged view of a portion A illustrated in FIG. 1.

[0017] FIG. 4 (a) is a plan view of a slit valve and FIG. 4 (b) is a sectional view along a line B-B illustrated in FIG. 4 (a).

[0018] FIG. 5 (a) is a view illustrating a state where a nozzle flow path is filled with a content liquid and no external force is applied to the container, FIG. 5 (b) is a view illustrating a state where a slit part of a slit valve is pushed and spread by the content liquid in the nozzle flow path, and FIG. 5 (c) is a view illustrating a state where the content liquid is discharged outside from the slit part.

[0019] FIG. 6 (a) is a view illustrating a state where the content liquid in the nozzle flow path starts returning to the container in a state where the external force to the container is canceled, FIG. 6 (b) is a view illustrating a state where air starts flowing into the nozzle flow path, and FIG. 6 (c) is a

view illustrating a state where the flow of the air into the nozzle flow path ends and the slit valve is closed.

[0020] FIG. 7 (a) is a plan view of a slit valve according to a first variation, and FIG. 7 (b) is a sectional view along a line C-C illustrated in FIG. 7 (a).

[0021] FIG. 8 is a sectional view of a slit valve according to a second variation.

[0022] FIG. 9 is a sectional view of a slit valve according to a third variation.

[0023] FIG. 10 is a sectional view of a slit valve according to a fourth variation.

[0024] FIG. 11 is a sectional view of a slit valve according to a fifth variation.

[0025] FIG. 12 is a sectional view of a discharge cap according to a modification of the first embodiment.

[0026] FIG. 13 (a) is a plan view illustrating a discharge cap according to the modification of the first embodiment in an open state, and FIG. 13 (b) is a sectional view along a line D-D illustrated in FIG. 13 (a).

[0027] FIG. 14 is a sectional view illustrating a closed state where a lid body is placed on a discharge cap according to a second embodiment.

[0028] FIG. 15 is an enlarged view of a portion E illustrated in FIG. 14.

[0029] FIG. 16 is a sectional view of a discharge cap according to a modification of the second embodiment.

## DESCRIPTION OF EMBODIMENTS

### First Embodiment

[0030] Embodiments of the present invention will be described below in detail with reference to the drawings. Constituent elements, types, combinations, shapes, and relative arrangements described in the embodiments are merely explanatory examples, and are not intended to limit the scope of the present invention solely thereto unless otherwise specified.

[0031] FIGS. 1 and 2 are sectional views illustrating a discharge cap 1A and a part of a container 3 according to a first embodiment. As illustrated in FIGS. 1 and 2, the discharge cap 1A according to the first embodiment includes a discharge nozzle 29 and is removably attached to a mouth part 15 of the container (a squeeze container) 3 having an elasticity. A liquid (a content liquid) CLq is stored in the interior space of the container 3. The discharge cap 1A discharges the content liquid CLq from the discharge nozzle 29 with a pressure increase in the interior space of the container 3 due to an external force applied to the container 3.

#### <Container 3>

[0032] The container 3 is molded of a synthetic resin such as polyethylene (PE) or polyethylene terephthalate (PET), and easily deforms and has an elasticity. The container 3 deforms so as to decrease the volume of the interior space when an external force is applied by compression (squeeze) or the like, and is restored to the original shape due to the elasticity when the external force is canceled.

[0033] The container 3 includes a body 13 for storing the content liquid CLq, and the mouth part 15 provided integrally with the body 13 and through which the content liquid CLq is taken in and out of the body 13. An external thread 17 for attaching the discharge cap 1A is formed on an outer circumferential portion of the mouth part 15.

[0034] The shape of the body 13 is not particularly limited thereto as long as the volume of the interior space is decreased when an external force is applied thereto and the body is restored to the original shape when the external force is canceled. For example, the body 13 can be formed in a bottomed cylindrical shape.

[0035] The content liquid CLq stored in the interior space of the container 3 is a viscous liquid such as a lubricant, a caulking agent, or honey. The viscosity of a lubricant is about 3100 mPa·s and the viscosity of honey is about 12000 mPa·s.

[0036] In the following descriptions, the content liquid CLq is, for example, a liquid having a viscosity of 500 mPa·s to 12000 mPa·s.

<Discharge Cap **1a** According to First Embodiment>

[0037] The discharge cap **1A** includes a cap body **5** that includes the discharge nozzle **29** and that is removably attached to the mouth part **15** of the container **3**, a slit valve **7A** (a valve body) and a holder **9** (a holding part) provided at a leading end part of the discharge nozzle **29**, and a lid body **11** removably attached to the cap body **5** to cover the discharge nozzle **29**.

[0038] The cap body **5** includes a ceiling wall **21** in a substantially discoid shape, a tubular outer wall part **23**, a tubular intermediate wall part **25**, and a tubular inner wall part **27**, which extend from the inner surface of the ceiling wall **21** toward the container **3** and have a substantially cylindrical shape, and the discharge nozzle **29** in a substantially cylindrical shape extending from a central portion of the outer surface of the ceiling wall **21** to the opposite side to the container **3**. In the present embodiment, the cap body **5** is made of a synthetic resin with such a hardness that does not deform at the time of discharge of the content liquid CLq.

[0039] The tubular outer wall part **23** extends from an outer edge part of the ceiling wall **21** to a region close to the body **13**. An annular extended part **31** extended outward in the radial direction is formed on the outer wall surface of an end part of the tubular outer wall part **23** (an end part on the side far from the ceiling wall **21**). The extended part **31** is provided to support the opening edge of the lid body **11** attached to the cap body **5**.

[0040] The tubular intermediate wall part **25** is provided on the side of the inner circumference of the tubular outer wall part **23** concentrically with the tubular outer wall part **23** at an appropriate distance therefrom. An end part of the tubular intermediate wall part **25** (an end part on the side far from the ceiling wall **21**) extends close to the base of the mouth part **15**. An internal thread **33** with which the external thread **17** of the mouth part **15** is removably fitted is formed on the inner wall surface of the tubular intermediate wall part **25**.

[0041] The tubular inner wall part **27** is provided on the side of the inner circumference of the tubular intermediate wall part **25** concentrically with the tubular intermediate wall part **25** at a distance necessary for the leading end portion of the mouth part **15** to be fitted therewith. Therefore, a space (a gap) **35** in which the mouth part **15** of the container **3** is inserted is formed between the tubular intermediate wall part **25** and the tubular inner wall part **27**.

[0042] The length of the tubular inner wall part **27** is smaller than that of the tubular intermediate wall part **25**. A taper having the diameter decreasing with the increasing distance from the ceiling wall **21** is provided on the outer circumferential surface of the tubular inner wall part **27**. The taper guides the mouth part **15** when the mouth part **15** is fitted to the cap body **5**. An outer circumferential surface of the tubular inner wall part **27** on the side of the ceiling wall **21** tightly adheres to the inner circumferential surface of the mouth part **15** to suppress leakage of the content liquid CLq in a state where the mouth part **15** is fitted to the cap body **5**.

<Discharge Nozzle **29**>

[0043] The discharge nozzle **29** has a tubular shape extending from the base end on a side near the ceiling wall **21** toward the leading end far from the ceiling wall **21**, and a nozzle flow path **29a** through which the content liquid CLq flows is provided inside the discharge nozzle **29**. The ceiling wall **21** is provided with an opening part **43** that causes the nozzle flow path **29a** to be communicated with the interior space of the container **3**. Therefore, the content liquid CLq flows so as to be capable of moving back and forth between the interior space of the container **3** and the nozzle flow path **29a**.

[0044] A sectional area SA**29a** of the nozzle flow path **29a** is smaller than an opening area OP**15** of the mouth part **15**. Accordingly, the thickness of the content liquid CLq discharged from the leading end of the discharge nozzle **29** can be thinned as compared to the content liquid CLq directly flowed out from the mouth part **15**. Furthermore, since the discharge nozzle **29** has a predetermined length, the content liquid CLq can be easily discharged to a desired place.

[0045] In the present embodiment, the discharge nozzle **29** has a cylindrical shape tapered from the base end on the side near the ceiling wall **21** toward the leading end. However, the discharge nozzle

**29** is not limited to this shape. For example, the sectional shape of the nozzle flow path **29a** may be uniform from the base end of the discharge nozzle **29** toward the leading end.

[0046] FIG. **3** is an enlarged view of a portion A illustrated in FIG. **1**. As illustrated in FIGS. **2** and **3**, an annular protruding part **39** protruded inward in the radial direction is formed on the inner circumference side of the leading end part of the discharge nozzle **29**.

[0047] The inner circumferential surface of the protruding part **39** is a discharge opening **37** on which a valve part **49** and a thin-walled tube part **51** included in the slit valve **7A** are arranged. The inner surface of the protruding part **39** (an annular face protruded inward in the radial direction in the nozzle flow path **29a**) is an abutting face **38** that abuts on a valve sidewall part **47** of the slit valve **7A** to prevent the slit valve **7A** from falling off.

[0048] A diameter expanded part **45** with a diameter expanded outward in the radial direction is formed on the outer circumferential surface of the leading end part of the discharge nozzle **29**. The diameter expanded part **45** is provided to removably fix the lid body **11**.

[0049] As illustrated in FIG. **3**, an annular concave part **41** is formed on the inner circumferential surface of the discharge nozzle **29** and at a position nearer the leading end in relation to the intermediate position of the discharge nozzle **29** in the length direction. The concave part **41** is provided to prevent the holder **9** from falling off the discharge nozzle **29**.

<Slit Valve **7a**>

[0050] FIG. **4 (a)** is a plan view of the slit valve **7A** and FIG. **4 (b)** is a sectional view along a line B-B illustrated in FIG. **4 (a)**. As illustrated in FIGS. **3**, **4 (a)**, and **4 (b)**, the slit valve **7A** includes the cylindrical valve sidewall part **47** (a held part), the substantially discoid valve part **49** from which the content liquid CLq in the nozzle flow path **29a** is discharged, and the thin-walled tube part **51** (a connecting part) that connects the valve sidewall part **47** and the valve part **49** to each other.

[0051] The slit valve **7A** is made of an elastic material softer than the discharge nozzle **29** (the cap body **5**). In the present embodiment, the slit valve **7A** is integrally molded of an elastic synthetic resin such as silicone having a hardness (a hardness measured by a durometer of type A) of about A20° to about A50°. Since being made by integral molding, the slit valve **7A** of the present embodiment is easily produced.

[0052] The valve sidewall part **47** is held on the discharge nozzle **29** by the holder **9**. A leading end face **47a** of the valve sidewall part **47** on a side near the thin-walled tube part **51** (more specifically, a face projecting outward in the radial direction relative to the thin-walled tube part **51**) abuts on the abutting face **38** of the protruding part **39** to prevent the slit valve **7A** from falling off the discharge nozzle **29**. Accordingly, a thickness TH47 of the valve sidewall part **47** is determined to a thickness required to prevent falling off the discharge nozzle **29**. In the present embodiment, the thickness TH47 of the valve sidewall part **47** is about 0.3 mm (millimeter) to about 1.0 mm.

[0053] The thin-walled tube part **51** connects an inner circumferential part of the valve sidewall part **47** on a side near the leading end face **47a** and an outer circumferential part of the valve part **49** on a side near an inner surface **54** to each other. The outside diameter of the thin-walled tube part **51** is substantially equal to the opening diameter of the discharge opening **37**. Accordingly, the outer circumferential surface of the thin-walled tube part **51** abuts on the discharge opening **37** in a state where the slit valve **7A** is held on the leading end part of the discharge nozzle **29**. In other words, the thin-walled tube part **51** is arranged so as to abut on the discharge opening **37** of the discharge nozzle **29**.

[0054] A thickness TH51 of the thin-walled tube part **51** is sufficiently smaller than the thickness TH47 of the valve sidewall part **47**. For example, the thickness TH51 of the thin-walled tube part **51** is equal to or less than a half (substantially a quarter to substantially a third) of the thickness TH47 of the valve sidewall part **47**. A length L51 of the thin-walled tube part **51** is about 0.3 mm to about 0.7 mm.

[0055] Accordingly, the area of the leading end face **47a** of the valve sidewall part **47** (in other

words, the area of a portion abutting on the abutting face **38** of the discharge nozzle **29**) can be sufficiently provided and falling of the slit valve **7A** from the discharge nozzle **29** can be effectively prevented.

[0056] The thin-walled tube part **51** more easily deforms than the valve sidewall part **47**. Even when the valve sidewall part **47** deforms associated with attachment of the slit valve **7A**, the thin-walled tube part **51** easily deforms and accordingly absorbs the deformation of the valve sidewall part **47**.

[0057] To specifically explain this configuration, the valve sidewall part **47** is sandwiched by the discharge nozzle **29** and the holder **9** from the outer circumference side and the inner circumference side as illustrated in FIG. **3**. Attachment of the slit valve **7A** to the discharge nozzle **29** is performed, for example, by relatively moving the holder **9** having the slit valve **7A** put on the leading end part toward the leading end of the discharge nozzle **29** through an inner void (that is, the nozzle flow path **29a**) of the discharge nozzle **29**. Therefore, at the time of attachment of the slit valve **7A**, the valve sidewall part **47** is sometimes deformed by friction between the inner circumferential surface of the discharge nozzle **29** and the outer circumferential surface of the holder **9**. Even when the valve sidewall part **47** is deformed, the thin-walled tube part **51** deforms itself to prevent the valve part **49** from being affected by the deformation.

[0058] There is also a case where the thin-walled tube part **51** deforms at the time of discharge of the content liquid CLq. For example, in a case where the content liquid CLq having a relatively high viscosity, such as a lubricant (viscosity: about 3100 mPa·s) or honey (viscosity: about 12000 mPa·s) is discharged, the thin-walled tube part **51** extends when an external force in the discharge direction is applied to the valve part **49** and contracts after the content liquid CLq is discharged. As will be described later, the thin-walled tube part **51** pulls an outer circumferential part **49a** toward the nozzle flow path **29a** at the time of contraction, and accordingly assists deformation of a slit part **49b** toward the nozzle flow path **29a**. As a result, the content liquid CLq after discharge is cut off in an improved manner.

<Valve Part **49**>

[0059] The outside diameter of the valve part **49** is substantially equal to the opening diameter of the discharge opening **37**. Accordingly, the outer circumferential surface of the valve part **49** is in contact with the discharge opening **37** in a state where the slit valve **7A** is held on the leading end part of the discharge nozzle **29**. In other words, the valve part **49** is arranged so as to be fitted in the discharge opening **37** of the discharge nozzle **29**. In the present embodiment, the outside diameter **049** of the valve part **49** is about 4 mm to about 7 mm.

[0060] An outer surface **53** of the valve part **49** is a concave face spherically depressed toward the side of the nozzle flow path **29a** (on the opposite side to the discharge direction of the content liquid CLq). Meanwhile, the inner surface **54** of the valve part **49** is a flat face orthogonal to a central axis line L of the cylinder of the valve sidewall part **47**.

[0061] Slits **55** extending outward in the radial direction from a center point O are formed on the valve part **49**. In the slit valve **7A**, the slits **55** are cross-shaped. The slits **55** are configured to open and close according to a difference between the pressure in the interior space of the container **3** and the atmospheric pressure. In the present embodiment, a length L<sub>55</sub> of each of the slits **55** is about 2 mm to about 5 mm.

[0062] As for the slits **55**, the intersecting angle between one slit and the other slit is not limited to 90 degrees. For example, the intersecting angle between one slit and the other slit may be changed in a range from 60 degrees to 90 degrees. The length of one slit and the length of the other slit may be different.

[0063] The valve part **49** includes the outer circumferential part **49a** coupled to the thin-walled tube part **51**, a slit part **49b** on which the slits **55** are provided, and an intermediate part **49c** located between the slit part **49b** and the outer circumferential part **49a**.

[0064] As described above, the outer surface **53** of the valve part **49** is a concave face in a spherical

shape and the inner surface **54** is a flat face. Therefore, the outer circumferential part **49a**, the intermediate part **49c**, and the slit part **49b** are formed to be gradually thinned from the outer circumference of the valve part **49** toward the center point O. Accordingly, the intermediate part **49c** is formed to be gradually thinned from the outer circumferential part **49a** toward the slit part **49b**.

[0065] In the present embodiment, the thickness of the slit part **49b** is equal to or less than a half of the thickness of the outer circumferential part **49a** and the slit part **49b** is provided on a side nearer the nozzle flow path **29a** in relation to an intermediate position TC**49** of the outer circumferential part **49a** in the thickness direction.

[0066] To cite an example, a largest thickness TH**49a** of the outer circumferential part **49a** is about 0.6 mm to about 0.7 mm and a largest thickness TH**49b** of the slit part **49b** is about 0.3 mm to about 0.35 mm. A smallest thickness of the slit part **49b** at the center point O is about 0.15 mm to about 0.25 mm.

[0067] With the configuration described above, at the time of discharge of the content liquid CLq, the slit part **49b** more easily deforms than the outer circumferential part **49a** due to a pressure change of the content liquid CLq (the interior space of the container **3**) and contributes to reduction of the diameter of the leading end side (downstream side) of the discharge nozzle **29**.

[0068] The outer circumferential part **49a** is thicker than the intermediate part **49c** and the slit part **49b**, and the intermediate part **49c** is thicker than the slit part **49b**. Accordingly, after discharge of the content liquid CLq, the outer circumferential part **49a** has a larger force (restoring force) to deform in a returning direction than the intermediate part **49c**, and the intermediate part **49c** has a larger restoring force than the slit part **49b**. As a result, the slit part **49b** is promptly deformed toward the nozzle flow path **29a** under the restoring forces from the outer circumferential part **49a** and the intermediate part **49c** in addition to its own restoring force, and improves cutoff of the content liquid CLq.

[0069] Furthermore, since the outer circumferential part **49a** and the intermediate part **49c** are thicker than the slit part **49b**, the both ends of each of the slits **55** are enabled to become less likely to be split.

[0070] Besides, since the slit valve **7A** is integrally molded of a synthetic resin such as silicone, the softness of the parts **49a**, **49b**, and **51** can be easily set by setting the thicknesses TH**49a**, TH**49b**, and TH**51** of the parts **49a**, **49b**, and **51**.

<Holder **9**>

[0071] As illustrated in FIG. **3**, the holder **9** is a hollow tubular body fitted in the inner circumferential part of the discharge nozzle **29** on the leading end side and is a substantially cylindrical member in the present embodiment. The holder **9** holds the valve sidewall part **47** along with the discharge nozzle **29** and prevents the slit valve **7A** from falling off the discharge nozzle **29**.

[0072] The holder **9** includes a small-diameter tube part **57** located on the side of the leading end of the discharge nozzle **29**, and a large-diameter tube part **59** located on a side nearer the base end of the discharge nozzle **29** in relation to the small-diameter tube part **57**. The base end of the small-diameter tube part **57** and the leading end of the large-diameter tube part **59** is integrally coupled to each other. An interior space **9a** of the holder **9** constitutes a part of the nozzle flow path **29a**.

[0073] The outside diameter of the small-diameter tube part **57** is substantially the same as the inside diameter of the valve sidewall part **47**. The outside diameter of the large-diameter tube part **59** is substantially the same as the inside diameter of the discharge nozzle **29**.

[0074] A flange part **61** protruded outward in the radial direction is provided on a base end portion of the outer circumferential surface of the large-diameter tube part **59**. The flange part **61** is fitted in the annular concave part **41** of the discharge nozzle **29**.

[0075] The holder **9** is positioned and is fixed to the discharge nozzle **29** by fitting of the flange part **61** into the concave part **41**.

[0076] When the holder **9** is fixed to the discharge nozzle **29**, the valve sidewall part **47** of the slit



valve **7A** is arranged between the discharge nozzle **29** and the small-diameter tube part **57**.

[0077] The leading end face **47a** of the valve sidewall part **47** abuts on the abutting face **38** of the protruding part **39**, and a base end face **47b** of the valve sidewall part **47** abuts on a leading end face **60** (a face with a diameter expanded outward in the radial direction relative to the small-diameter tube part **57**) of the large-diameter tube part **59**.

[0078] As described above, the valve sidewall part **47** is held in a space formed by the discharge nozzle **29** and the holder **9**. As a result, falling of the slit valve **7A** from the discharge nozzle **29** is prevented. The thin-walled tube part **51** extends from a gap between the small-diameter tube part **57** and the protruding part **39** to the side of the leading end of the discharge nozzle **29**.

<Lid Body **11**>

[0079] As illustrated in FIGS. **1** and **3**, the lid body **11** is a member having a bottomed cylindrical shape including a cylindrical wall part **65** in a cylindrical shape, and a top board **67** in a discoid shape. A cylindrical latch part **69** is provided in a protruded manner at a central portion of an inner surface **67a** of the top board **67**.

[0080] The leading end portion of the discharge nozzle **29** is inserted into the interior space of the latch part **69**. On the inner circumferential surface of an open end part of the latch part **69**, a taper expanding toward the open end is formed. This taper is provided to guide the leading end portion of the discharge nozzle **29**.

[0081] An abutting part **71** raised in a spherical shape toward the open end of the latch part **69** is provided at a portion of the top board **67** on an inner circumference side than the latch part **69**. The surface of the abutting part **71** is brought to contact with the outer surface **53** of the valve part **49** to enhance the sealing property of the valve part **49** at the time of attachment of the lid body **11** to the cap body **5**.

<Operation at Time of Discharging Content Liquid CLq>

[0082] An operation at the time of discharging a content liquid CLq is explained next.

[0083] FIG. **5(a)** is a view illustrating a state where the nozzle flow path **29a** is filled with the content liquid CLq and no external force is applied to the container **3**, FIG. **5 (b)** is a view illustrating a state where the slit part **49b** of the slit valve **7A** is pushed and spread by the content liquid CLq in the nozzle flow path **29a**, and FIG. **5 (c)** is a view illustrating a state where the content liquid CLq is discharged outside from the slit part **49b**.

[0084] FIGS. **5 (a)** to **5 (c)** illustrate the leading end part of the discharge nozzle **29**. In these drawings, the leading end part of the discharge nozzle **29** faces downward.

[0085] An operation in a case where an external force is applied to the container **3** is explained first. As illustrated in FIG. **5 (a)**, the slit part **49b** of the slit valve **7A** is in a closed state in a state where the nozzle flow path **29a** is filled with the content liquid CLq and no external force is applied to the container **3**.

[0086] Therefore, the content liquid CLq in the nozzle flow path **29a** is kept in the nozzle flow path **29a** by the slit valve **7A** (the valve part **49**) in a closed state.

[0087] As illustrated in FIG. **5 (b)**, when an external force is applied to the container **3** by compression or the like, the pressure in the interior space is increased and a force PR+ in the discharge direction is applied to the content liquid CLq in the nozzle flow path **29a**. The force PR+ in the discharge direction is proportional to a difference between the pressure in the interior space of the container **3** and the atmospheric pressure. The force PR+ in the discharge direction is transmitted through the content liquid CLq and acts on the inner surface **54** of the valve part **49**.

[0088] The slit part **49b** is deformed in the discharge direction by the content liquid CLq. With the deformation of the slit part **49b**, a gap (a discharge port) is formed between the slits **55**.

[0089] Since being smaller in the thickness than the outer circumferential part **49a** and the intermediate part **49c**, the slit part **49b** deforms more largely than the outer circumferential part **49a** and the intermediate part **49c**. Furthermore, since being smaller in the thickness than the outer circumferential part **49a**, the intermediate part **49c** deforms more largely than the outer

circumferential part **49a**. Accordingly, the inner surface **54** of the valve part **49** forms an inclined face toward the discharge port and smoothly guides the content liquid CLq.

[0090] Subsequently, as illustrated in FIG. 5 (c), the content liquid CLq is continuously discharged from the gap (the discharge port) between the slits **55**. In this discharge cap **1A**, the thickness of the slit part **49b** is smaller than those of the outer circumferential part **49a** and the intermediate part **49c**. Accordingly, the slit part **49b** deforms more largely with a pressure change of the content liquid CLq (a pressure change in the interior space of the container **3**) than the outer circumferential part **49a** and the intermediate part **49c**.

[0091] As a result, the content liquid CLq can be discharged from the discharge nozzle **29** without application of an excessive external force to the container **3**. Furthermore, since the thickness of the content liquid CLq discharged from the discharge nozzle **29** can be limited by the size of the slit part **49b** (the slits **55**), an inconvenience that the content liquid CLq becomes too thick is suppressed.

[0092] An operation at the time of sucking the content liquid CLq into the discharge nozzle **29** is explained next.

[0093] FIG. 6 (a) is a view illustrating a state where the content liquid CLq in the nozzle flow path **29a** starts returning to the container **3** in a state where the external force to the container **3** is canceled, FIG. 6 (b) is a view illustrating a state where air AR starts flowing into the nozzle flow path **29a**, and FIG. 6 (c) is a view illustrating a state where the flow of the air AR into the nozzle flow path **29a** ends and the slit valve **7A** is closed.

[0094] Also in FIGS. 6 (a) to 6 (c), the leading end part of the discharge nozzle **29** faces downward similarly in FIGS. 5 (a) to 5 (c).

[0095] As illustrated in FIG. 6 (a), when the external force to the container **3** is canceled after discharge of the content liquid CLq, a force PR<sub>-</sub> (hereinafter, referred to as “force PR<sub>-</sub> in the returning direction”) to return to the container **3** is applied to the content liquid CLq in the nozzle flow path **29a**. The force PR<sub>-</sub> in the returning direction is generated because the container **3** in the contracted state is expanding to the original shape with its elasticity.

[0096] Besides, the outer circumferential part **49a**, the intermediate part **49c**, and the slit part **49b** are also restoring to the original shapes with their elasticities. Since being larger in the thickness than the intermediate part **49c** and the slit part **49b**, the outer circumferential part **49a** has a larger restoring force than the intermediate part **49c** and the slit part **49b**. Similarly, the intermediate part **49c** has a larger restoring force than the slit part **49b**.

[0097] As a result, after discharge of the content liquid CLq, the slit part **49b** is promptly deformed toward the nozzle flow path **29a** under the restoring forces of the outer circumferential part **49a** and the intermediate part **49c** in addition to its own restoring force.

[0098] As illustrated in FIG. 6 (b), after the content liquid CLq passes through the valve part **49**, an amount of air AR compensating the discharged content liquid CLq flows in the nozzle flow path **29a**. At the time of inflow of the air AR, the slit part **49b** deforms toward the nozzle flow path **29a** to form a gap between the slits **55**.

[0099] The thickness of the slit part **49b** is smaller than those of the outer circumferential part **49a** and the intermediate part **49c**. Accordingly, the slit part **49b** deforms more largely than the outer circumferential part **49a** and the intermediate part **49c**. With the deformation of the slit part **49b**, the gap between the slits **55** can have a sufficient size and the air AR can be efficiently taken into the nozzle flow path **29a**.

[0100] The thin-walled tube part **51** formed to have a thin wall is sometimes inclined in a circumferentially inward direction of the thin-walled tube part **51**. For example, the thin-walled tube part **51** is sometimes inclined in the circumferentially inward direction when the amount of air AR taken into the nozzle flow path **29a** is sufficiently large, when the speed at which the air AR is taken in is sufficiently high, or when the viscosity of the content liquid CLq is high. When the thin-walled tube part **51** is inclined, the slit part **49b** is largely inclined toward the nozzle flow path **29a**

correspondingly to the inclination of the thin-walled tube part **51**. Accordingly, the gap between the slits **55** becomes larger than in a case where the thin-walled tube part **51** is not inclined. As a result, also with the inclination of the thin-walled tube part **51**, the air AR can be efficiently taken into the nozzle flow path **29a**.

[0101] Subsequently, as illustrated in FIG. **6** (c), the force PR<sup>-</sup> in the returning direction to the content liquid CLq is lost with restoration of the container **3** to the original shape. With the loss of the force PR<sup>-</sup> in the returning direction, the slit part **49b** of the slit valve **7A** is brought to the closed state.

[0102] Therefore, the content liquid CLq in the nozzle flow path **29a** is kept in the nozzle flow path **29a** by the slit valve **7A** (the valve part **49**) in a closed state.

[0103] As is apparent from the above descriptions, the slit part **49b** deforms according to a pressure change of the content liquid CLq (a pressure change in the interior space of the container **3**).

Therefore, the gap between the slits **55** can be made sufficiently large by the force PR<sup>-</sup> in the returning direction. Accordingly, the gap is enabled to have a size required to efficiently suck the content liquid CLq and air AR into the nozzle flow path **29a**, and the content liquid CLq can be cut off in an improved manner.

[0104] In the slit valve **7A**, since the outer circumferential part **49a** can be formed to have a sufficient thickness, the restoring force of the outer circumferential part **49a** after discharge of the content liquid CLq can be enhanced. Similarly, the restoring force of the intermediate part **49c** also can be enhanced as compared to the slit part **49b**.

[0105] The force PR<sup>+</sup> in the discharge direction applied to the valve part **49** (particularly, to the inner surface **54** of the valve part **49**) at the time of discharge of the content liquid CLq is increased as the viscosity of the content liquid CLq is increased. Since being thinner and more easily deforming than the outer circumferential part **49a** of the valve part **49**, the thin-walled tube part **51** is sometimes extended due to the force PR<sup>+</sup> in the discharge direction applied to the outer circumferential part **49a** as indicated by an arrow in FIG. **5** (b).

[0106] When the external force to the container **3** is canceled while the thin-walled tube part **51** remains extended, the thin-walled tube part **51** starts contracting in the return direction with its elasticity as illustrated by an arrow in FIG. **6** (a). The outer circumferential part **49a** is pulled toward the container **3** by contraction of the thin-walled tube part **51**. The contractive force of the thin-walled tube part **51** assists deformation of the slit part **49b** toward the nozzle flow path **29a** after discharge of the content liquid CLq.

[0107] As described above, there is a case where the content liquid CLq is enabled to be cut off in a further improved manner also by extension and contraction of the thin-walled tube part **51**.

[0108] The discharge nozzle **29** of the discharge cap **1A** has a cylindrical shape tapered from the base end (the upstream side) toward the leading end (the downstream side). Therefore, the content liquid can be easily discharged into a recess. As a result, the content liquid CLq can be easily discharged to various places.

#### <Variations of Slit Valve>

[0109] Variations of the slit valve are explained below.

[0110] FIG. **7** (a) is a plan view of a slit valve **7B** according to a first variation, FIG. **7** (b) is a sectional view along a line C-C illustrated in FIG. **7** (a), FIG. **8** is a sectional view of a slit valve **7C** according to a second variation, FIG. **9** is a sectional view of a slit valve **7D** according to a third variation, FIG. **10** is a sectional view of a slit valve **7E** according to a fourth variation, and FIG. **11** is a sectional view of a slit valve **7F** according to a fifth variation.

#### <Slit Valve **7b** According to First Variation>

[0111] The slit valve **7B** according to the first variation illustrated in FIGS. **7** (a) and **7** (b) is characterized in that a slit **55B** has a straight line shape, in other words, the slit **55B** is formed of one linear slit.

[0112] While being different from the slit valve **7A** described above in that the slit **55B** is formed of

one linear slit, the slit valve 7B is the same as the slit valve 7A in the rest of the configuration.

[0113] Therefore, also with use of the slit valve 7B in the discharge cap 1A, the slit part 49b deforms according to a pressure change of the content liquid CLq. Therefore, an external force applied to the container 3 to discharge the content liquid CLq can be suppressed to an appropriate magnitude and the content liquid CLq discharged from the discharge nozzle 29 does not become too thick.

[0114] In the slit valve 7B, the thickness of the slit part 49b is equal to or less than a half of the thickness of the outer circumferential part 49a and the slit part 49b is provided on a side nearer the nozzle flow path 29a in relation to the intermediate position TC49 of the outer circumferential part 49a in the thickness direction.

[0115] Therefore, also with use of the slit valve 7B in the discharge cap 1A, the restoring force of the outer circumferential part 49a after discharge of the content liquid CLq can be enhanced.

[0116] With the configuration described above, the slit part 49b is promptly deformed toward the nozzle flow path 29a after discharge of the content liquid CLq. As a result, the content liquid CLq can be cut off in an improved manner. Also in the slit valve 7B, when the viscosity of the content liquid CLq is sufficiently high, the outer circumferential part 49a is pulled toward the container 3 by contraction of the thin-walled tube part 51 and accordingly the content liquid CLq can be cut off in an improved manner.

<Slit valve 7C according to second variation>

[0117] The slit valve 7C according to the second variation illustrated in FIG. 8 is characterized in that an outer surface 53C of the valve part 49 is a flat face and that an inner surface 54C is a concave face spherically depressed in the discharge direction of the content liquid CLq.

[0118] Also with use of the slit valve 7C in the discharge cap 1A, the slit part 49b deforms according to a pressure change of the content liquid CLq. Therefore, an external force applied to the container 3 to discharge the content liquid CLq can be suppressed to an appropriate magnitude and the content liquid CLq discharged from the discharge nozzle 29 does not become too thick.

[0119] The outer circumferential part 49a and the intermediate part 49c promptly deform the slit part 49b toward the nozzle flow path 29a after discharge of the content liquid CLq. As a result, the content liquid CLq can be cut off in an improved manner. Furthermore, when the viscosity of the content liquid CLq is sufficiently high, the content liquid CLq can be cut off in a further improved manner by extension and contraction of the thin-walled tube part 51.

<Slit Valve 7d According to Third Variation>

[0120] The slit valve 7D according to the third variation illustrated in FIG. 9 is characterized in that each of an outer surface 53D and an inner surface 54D of the valve part 49 is a concave face spherically depressed.

[0121] Also with use of the slit valve 7D in the discharge cap 1A, the slit part 49b deforms according to a pressure change of the content liquid CLq. Therefore, an external force applied to the container 3 to discharge the content liquid CLq can be suppressed to an appropriate magnitude and the content liquid CLq discharged from the discharge nozzle 29 does not become too thick.

[0122] The outer circumferential part 49a and the intermediate part 49c promptly deform the slit part 49b toward the nozzle flow path 29a after discharge of the content liquid CLq. As a result, the content liquid CLq can be cut off in an improved manner. Furthermore, when the viscosity of the content liquid CLq is sufficiently high, the content liquid CLq can be cut off in a further improved manner by extension and contraction of the thin-walled tube part 51.

<Slit Valve 7b According to Fourth Variation>

[0123] The slit valve 7E according to the fourth variation illustrated in FIG. 10 is characterized in that the thickness of the intermediate part 49c of the valve part 49 is stepwise thinned toward the center of the valve part 49 and that the slit part 49b is provided on a side nearer the nozzle flow path 29a in relation to the intermediate position TC49 of the outer circumferential part 49a in the thickness direction.

[0124] Also with use of the slit valve 7E in the discharge cap 1A, the slit part 49b deforms according to a pressure change of the content liquid CLq. Therefore, an external force applied to the container 3 to discharge the content liquid CLq can be suppressed to an appropriate magnitude and the content liquid CLq discharged from the discharge nozzle 29 does not become too thick. [0125] Furthermore, the thickness of the slit part 49b is equal to or less than a half of the thickness of the outer circumferential part 49a and the slit part 49b is provided on the side nearer the nozzle flow path 29a in relation to the intermediate position TC49. Accordingly, after discharge of the content liquid CLq, the slit part 49b is promptly deformed toward the nozzle flow path 29a. As a result, the content liquid CLq can be cut off in an improved manner after discharge of the content liquid CLq.

<Slit Valve 7F According to Fifth Variation>

[0126] The slit valve 7F according to the fifth variation illustrated in FIG. 11 is characterized in that a portion of an outer surface 53F of the valve part 49 corresponding to the intermediate part 49c and a portion of an inner surface 54F corresponding to the intermediate part 49c are tapered faces inclined toward the slit part 49b, and that the slit part 49b is provided on a side nearer the nozzle flow path 29a in relation to the intermediate position TC49 of the outer circumferential part 49a in the thickness direction.

[0127] Also with use of the slit valve 7F in the discharge cap 1A, the slit part 49b deforms according to a pressure change of the content liquid CLq. Therefore, an external force applied to the container 3 to discharge the content liquid CLq can be suppressed to an appropriate magnitude and the content liquid CLq discharged from the discharge nozzle 29 does not become too thick.

[0128] Furthermore, the thickness of the slit part 49b is equal to or less than a half of the thickness of the outer circumferential part 49a and the slit part 49b is provided on the side nearer the nozzle flow path 29a in relation to the intermediate position TC49. Accordingly, after discharge of the content liquid CLq, the slit part 49b is promptly deformed toward the nozzle flow path 29a. As a result, the content liquid CLq can be cut off in an improved manner after discharge of the content liquid CLq.

<Discharge Cap 1B According to Modification of First Embodiment>

[0129] FIG. 12 is a sectional view of a discharge cap 1B according to a modification of the first embodiment, FIG. 13 (a) is a plan view illustrating the discharge cap 1B according to the modification of the first embodiment in an open state, and FIG. 13 (b) is a sectional view along a line D-D illustrated in FIG. 13 (a).

[0130] As illustrated in FIGS. 12, 13 (a), and 13 (b), the discharge cap 1B according to the modification of the first embodiment is characterized in that the discharge nozzle 29 is shorter than the discharge cap 1A of the first embodiment. Configurations identical to those of the discharge cap 1A according to the first embodiment are denoted by like reference signs and explanations thereof are omitted.

[0131] The lid body 11 is attached to the cap body 5 of the discharge cap 1B with a hinge 75. A projection 79 is provided on the lid body 11 on the opposite side to the hinge 75. A concave part 77 is provided on the cap body 5 on the opposite side to the hinge 75. The projection 79 is provided to be caught by a finger or a nail of a user in a closed state (see FIG. 12) where the lid body 11 is placed on the cap body 5. The concave part 77 is provided to enable a finger or the like of a user to easily catch the projection 79.

[0132] As described above, also in the discharge cap 1B according to the modification having the shorter discharge nozzle 29, the content liquid CLq can be discharged from the discharge nozzle 29 without application of an excessively large external force to the container 3. Also in the discharge cap 1B, the diameter of the leading end part of the discharge nozzle 29 can be decreased to enable a small amount of the content liquid CLq to be discharged.

<Discharge Cap 1C According to Second Embodiment>

[0133] FIG. 14 is a sectional view illustrating a closed state where the lid body 11 is placed on a

discharge cap **1C** according to a second embodiment, and FIG. **15** is an enlarged view of a portion E illustrated in FIG. **14**.

[0134] As illustrated in FIGS. **14** and **15**, the discharge cap **1C** according to the second embodiment is characterized in that a holder **9C** is attached to the outer circumferential surface of the leading end part of a discharge nozzle **29C** as compared to the discharge cap **1A** according to the first embodiment. Configurations identical to those of the discharge cap **1A** according to the first embodiment are denoted by like reference signs and explanations thereof are omitted.

[0135] The discharge nozzle **29C** includes a large-diameter cylindrical wall **81** having the largest diameter, an intermediate-diameter cylindrical wall **83** smaller in the diameter than the large-diameter cylindrical wall **81**, and a small-diameter cylindrical wall **85** smaller in the diameter than the intermediate-diameter cylindrical wall **83** arranged in this order from the side of the ceiling wall **21** toward the leading end side. In the present embodiment, the large-diameter cylindrical wall **81** is the longest, and accounts for about three-fourths of the length of the discharge nozzle **29C**. The intermediate-diameter cylindrical wall **83** is the second longest and the small-diameter cylindrical wall **85** is the shortest. Therefore, it can be said that the discharge nozzle **29C** is stepwise reduced in the outside diameter at a portion on the leading end side.

[0136] The outside diameter of the small-diameter cylindrical wall **85** is substantially the same as the inside diameter of the valve sidewall part **47** of the slit valve **7A**. An annular projecting part **87** is formed on the outer circumferential surface of the intermediate-diameter cylindrical wall **83**. The projecting part **87** is provided to prevent falling of the holder **9C**.

[0137] The holder **9C** is a substantially cylindrical member and the inside diameter of the holder **9C** is configured to be substantially the same as the outside diameter of the intermediate-diameter cylindrical wall **83** except for a leading end portion thereof. In the present embodiment, the outside diameter of the intermediate-diameter cylindrical wall **83** is configured to be substantially the same as the outside diameter of the cylindrical valve sidewall part **47** of the slit valve **7A**. The holder **9C** is fitted with the intermediate-diameter cylindrical wall **83** of the discharge nozzle **29C**. In a state where the holder **9C** is fitted therewith, the small-diameter cylindrical wall **85** is arranged in the interior space of the holder **9C**.

[0138] On the inner circumference side of the leading end part of the holder **9C**, an annular protruding part **89** protruded inward in the radial direction is formed. The protruding part **89** is provided for the same purpose as that of the protruding part **39** included in the discharge cap **1A** of the first embodiment. A diameter expanded part **93** that engages with the latch part **69** of the lid body **11** is provided on the outer circumferential surface of the leading end part of the holder **9C**.

[0139] An annular concave part **91** is formed on the inner circumferential surface of the holder **9C** at a position near the base end. The concave part **91** fits with the projecting part **87** provided on the outer circumferential surface of the intermediate-diameter cylindrical wall **83** and prevents falling of the holder **9C** from the discharge nozzle **29C**.

[0140] According to the configuration described above, in the discharge cap **1C**, attachment of the holder **9C** to the discharge nozzle **29C** enables the cylindrical valve sidewall part **47** of the slit valve **7A** to be held between the inner circumferential surface of the holder **9C** and the outer circumferential surface of the small-diameter cylindrical wall **85** of the discharge nozzle **29C**, and enables the valve part **49** to be held by the inner circumferential portion of the annular protruding part **89**.

[0141] The discharge cap **1C** according to the second embodiment also exerts identical operational effects as those of the discharge cap **1A** according to the first embodiment.

<Discharge Cap **1D** According to Modification of Second Embodiment>

[0142] FIG. **16** is a sectional view of a discharge cap **1D** according to a modification of the second embodiment. As illustrated in FIG. **16**, the discharge cap **1D** according to the modification of the second embodiment is obtained by applying the configurations of the discharge nozzle **29C** and the holder **9C** of the second embodiment to the discharge cap **1B** (see FIG. **12** and the like) according

to the modification of the first embodiment. Configurations identical to those of the discharge cap **1B** according to the modification of the first embodiment and the discharge cap **1C** according to the second embodiment are denoted by like reference signs and explanations thereof are omitted.

[0143] The discharge cap **1D** according to the modification of the second embodiment also exerts identical operational effects as those of the discharge cap **1A** according to the first embodiment.

<Other Modifications>

[0144] While the discharge caps **1A** and **1C** described above have a configuration in which the lid body **11** is fitted with the cap body **5**, the discharge caps **1A** and **1C** are not limited to this configuration. For example, it is possible to form an external thread on the outer circumferential surface of the tubular outer wall part **23** of the cap body **5** and to form an internal thread on the inner circumferential surface of the cylindrical wall part **65** of the lid body **11**.

[0145] While the discharge caps **1B** and **1D** described above have a configuration in which the lid body **11** and the cap body **5** are connected to each other with the hinge **75**, the discharge caps **1B** and **1C** are not limited to this configuration. For example, a configuration in which the lid body **11** is fitted with the cap body **5** without connecting the lid body **11** and the cap body **5** to each other may be adopted.

[0146] In the discharge caps **1A**, **1B**, **1C**, and **1D** described above, the lengths of the discharge nozzles **29** and **29A** can be appropriately determined.

Summary of Aspect Examples, Actions, and Effects of Present Invention

<First Aspect>

[0147] The present aspect provides the discharge cap **1A**, **1B**, **1C**, or **1D** that is attached to the mouth part **15** of the container **3** having an elasticity and that discharges a content liquid CLq associated with a pressure increase in an interior space of the container **3** due to an external force applied to the container **3**, the discharge cap comprising: the discharge nozzle **29** or **29C** that forms the nozzle flow path **29a** with the sectional area SA**29a** smaller than the opening area OP**15** of the mouth part **15**; and the slit valve **7A**, **7B**, **7C**, **7D**, **7E**, or **7F** that is provided on the discharge nozzle **29** or **29C** and that is made of an elastic material softer than the discharge nozzle **29** or **29C**, in which the slit valve **7A**, **7B**, **7C**, **7D**, **7E**, or **7F** includes the valve part **49** through which the content liquid CLq in the nozzle flow path **29a** is discharged, the tubular valve sidewall part **47** (held part) that is held on the discharge nozzle **29** or **29C** by the holder **9** or **9C**, and the tubular thin-walled tube part **51** (a connecting part) that connects the valve part **49** and the valve sidewall part **47** to each other, the valve part **49** includes the slit part **49b** provided with the slit **55** or **55B** opening and closing according to a difference between an atmospheric pressure and a pressure in the interior space of the container **3**, and the outer circumferential part **49a** coupled to the thin-walled tube part **51**, and the slit part **49b** more easily deforms with a pressure change in the interior space than the outer circumferential part **49a**, deforms in a discharge direction of the content liquid CLq at a time of discharge of the content liquid CLq, and deforms toward the nozzle flow path **29a** after the content liquid CLq is discharged.

[0148] With the discharge cap **1A**, **1B**, **1C**, or **1D** according to the present aspect, the slit part **49b** deforms with a pressure change in the interior space, and the slits **55** or the slit **55B** opens to form a discharge port for the content liquid CLq. Since the slit part **49b** more easily deforms than the outer circumferential part **49a**, an external force applied to the container **3** to discharge the content liquid CLq can be suppressed to an appropriate magnitude and an inconvenience that the content liquid CLq discharged from the discharge nozzle **29** or **29C** becomes too thick is suppressed.

[0149] Furthermore, since the outer circumferential part **49a** is less likely to deform than the slit part **49b**, the restoring force thereof after the deformation is larger than that of the slit part **49b**. Accordingly, after discharge of the content liquid CLq, the restoring force of the outer circumferential part **49a** can be increased. The restoring force of the outer circumferential part **49a** assists the restoring force of the slit part **49b** to return to the closed state. As a result, the content liquid CLq after discharge can be cut off in an improved manner.

<Second Aspect>

[0150] In the discharge cap **1A**, **1B**, **1C**, or **1D** according to the present aspect, the slit part **49b** of the slit valve **7A**, **7B**, **7C**, **7D**, **7E**, or **7F** is integrally molded of a same elastic material as that of the outer circumferential part **49a**, and the thickness **TH49b** of the slit part **49b** and the thickness **TH51** of the thin-walled tube part **51** are smaller than the thickness **TH49a** of the outer circumferential part **49a**.

[0151] With the discharge cap **1A**, **1B**, **1C**, or **1D** according to the present aspect, the slit valve **7A**, **7B**, **7C**, **7D**, **7E**, or **7F** is easily produced and softness thereof can be managed by the thicknesses **TH49a** and **TH49b** of the parts **49a** and **49b**.

<Third Aspect>

[0152] In the discharge cap **1A**, **1B**, **1C**, or **1D** according to the present aspect, the valve part **49** of the slit valve **7A**, **7B**, **7C**, **7D**, **7E**, or **7F** includes the intermediate part **49c** between the slit part **49b** and the outer circumferential part **49a**, and the intermediate part **49c** is gradually thinned from the outer circumferential part **49a** to the slit part **49b**.

[0153] With the discharge cap **1A**, **1B**, **1C**, or **1D** according to the present aspect, after discharge of the content liquid **CLq**, the outer circumferential part **49a** has a larger restoring force than the intermediate part **49c** and the intermediate part **49c** has a larger restoring force than the slit part **49b**. As a result, the slit part **49b** promptly deforms toward the nozzle flow path **29a** under the restoring forces from the outer circumferential part **49a** and the intermediate part **49c** in addition to its own restoring force, and cuts off the content liquid **CLq** in an improved manner.

<Fourth Aspect>

[0154] In the discharge cap **1A**, **1B**, **1C**, or **1D** according to the present aspect, the valve part **49** of the slit valve **7A** or **7B** has the outer surface **53** being a spherical concave face depressed toward the nozzle flow path **29a**, and the inner surface **54** being a flat face.

[0155] With the discharge cap **1A**, **1B**, **1C**, or **1D** according to the present aspect, the amount of deformation of the outer circumferential part **49a** caused by a pressure from the content liquid **CLq** applied to the inner surface **54** of the outer circumferential part **49a** can be increased. Accordingly, after discharge of the content liquid **CLq**, a force of the outer circumferential part **49a** to deform in the returning direction can be increased and the slit part **49b** can be promptly deformed toward the nozzle flow path **29a**. As a result, the content liquid **CLq** can be cut off in an improved manner.

<Fifth Aspect>

[0156] In the discharge cap **1A**, **1B**, **1C**, or **1D** according to the present aspect, the slit part **49b** in the valve part **49** of the slit valve **7A**, **7B**, **7E**, or **7F** has a thickness equal to or less than a half of a thickness of the outer circumferential part **49a**, and the slit part **49b** is located on a side nearer the nozzle flow path **29a** in relation to the intermediate position **TC49** of the outer circumferential part **49a** in a thickness direction.

[0157] With the discharge cap **1A**, **1B**, **1C**, or **1D** according to the present aspect, the slit part **49b** is provided to be shifted toward the nozzle flow path **29a** in relation to the intermediate position **TC49**. Therefore, as compared to a configuration in which the slit part **49b** is provided to be shifted toward the discharge side in relation to the intermediate position **TC49**, the amount of deformation of the slit part **49b** at the time of discharge of the content liquid **CLq** can be increased. Therefore, the restoring force of the slit part **49b** after discharge of the content liquid **CLq** is enhanced. As a result, the slit part **49b** can be promptly deformed toward the nozzle flow path **29a** and can cut off the content liquid **CLq** in an improved manner.

<Sixth Aspect>

[0158] In the discharge cap **1A**, **1B**, **1C**, or **1D** according to the present aspect, the thin-walled tube part **51** (the connecting part) of the slit valve **7A**, **7B**, **7C**, **7D**, **7E**, or **7F** more easily deforms with a pressure change in the interior space of the container **3** than the outer circumferential part **49a**, extends in the discharge direction at a time of discharge of the content liquid **CLq**, and contracts in a returning direction after the content liquid **CLq** is discharged.



[0159] With the discharge cap **1A**, **1B**, **1C**, or **1D** according to the present aspect, after the content liquid CLq is discharged, the outer circumferential part **49a** can be pulled toward the nozzle flow path **29a** with a contractive force of the thin-walled tube part **51**. Therefore, the slit part **49b** can be promptly deformed toward the nozzle flow path **29a**. As a result, the content liquid CLq can be cut off in an improved manner.

#### <Seventh Aspect>

[0160] In the discharge cap **1A**, **1B**, **1C**, or **1D** according to the present aspect, the thin-walled tube part **51** (the connecting part) of the slit valve **7A**, **7B**, **7C**, **7D**, **7E**, or **7F** more easily deforms with a pressure change in the interior space of the container **3** than the outer circumferential part **49a**, and is inclined in a circumferentially inward direction of the thin-walled tube part **51** after the content liquid CLq is discharged.

[0161] With the discharge cap **1A**, **1B**, **1C**, or **1D** according to the present aspect, the gap between the slits **55** can be enlarged after discharge of the content liquid CLq, and air AR can be efficiently drawn into the nozzle flow path **29a**.

#### REFERENCE SIGNS LIST

[0162] **1A** . . . discharge cap according to first embodiment, **1B** . . . discharge cap according to modification of first embodiment, **1C** . . . discharge cap according to second embodiment, **1D** . . . discharge cap according to modification of second embodiment, **3** . . . container, **5** . . . cap body, **7A** . . . slit valve, **7B** . . . slit valve according to first variation, **7C** . . . slit valve according to second variation, **7D** . . . slit valve according to third variation, **7E** . . . slit valve according to fourth variation, **7F** . . . slit valve according to fifth variation, **9** . . . holder, **9C** . . . holder, **9a** . . . interior space of holder, **11** . . . lid body, **13** . . . body, **15** . . . mouth part, **17** . . . external thread, **21** . . . ceiling wall, **23** . . . tubular outer wall part, **25** . . . tubular intermediate wall part, **27** . . . tubular inner wall part, **29** . . . discharge nozzle, **29C** . . . discharge nozzle, **29a** . . . nozzle flow path, **31** . . . extended part, **33** . . . internal thread, **35** . . . space (gap), **37** . . . discharge opening, **38** . . . abutting face, **39** . . . protruding part, **41** . . . concave part, **43** . . . opening part, **45** . . . diameter expanded part, **47** . . . valve sidewall part, **47a** . . . leading end face of valve sidewall part, **49** . . . valve part, **49a** . . . outer circumferential part of valve part, **49b** . . . slit part of valve part, **49c** . . . intermediate part of valve part, **51** . . . thin-walled tube part, **53** . . . outer surface of valve part, **53C** . . . outer surface of valve part, **53D** . . . outer surface of valve part, **54** . . . inner surface of valve part, **54C** . . . inner surface of valve part, **54D** . . . inner surface of valve part, **55** . . . cross-shaped slit, **55B** . . . straight-line shaped slit, **57** . . . small-diameter tube part of holder, **59** . . . large-diameter tube part of holder, **60** . . . leading end face of large-diameter tube part, **61** . . . flange part of holder, **65** . . . cylindrical wall part of lid body, **67** . . . top board of lid body, **67a** . . . inner surface of top board, **69** . . . latch part of lid body, **71** . . . abutting part of lid body, **75** . . . hinge, **77** . . . concave part of cap body, **79** . . . projection of lid body, **81** . . . large-diameter cylindrical wall of discharge nozzle, **83** . . . intermediate-diameter cylindrical wall of discharge nozzle, **85** . . . small-diameter cylindrical wall of discharge nozzle, **87** . . . projecting part of intermediate-diameter cylindrical wall, **89** . . . protruding part of holder, **91** . . . concave part of holder, CLq . . . content liquid, OP**15** . . . opening area of mouth part, SA**29a** . . . sectional area of nozzle flow path, TH**47** . . . thickness of valve sidewall part,  $\Phi$ **49** . . . outside diameter of valve part, TC**49** . . . intermediate position of outer circumferential part in thickness direction, TH**49a** . . . largest thickness of outer circumferential part, TH**49b** . . . largest thickness of slit part, TH**51** . . . thickness of thin-walled tube part, L**51** . . . length of thin-walled tube part, L**55** . . . length of slit, O . . . center point of valve part, L . . . central axis line of valve sidewall part, PR+ . . . force in discharge direction, PR- . . . force in returning direction, AR . . . air

## Claims

1. A discharge cap that is attached to a mouth part of a container having an elasticity and that discharges a content liquid associated with a pressure increase in an interior space of the container due to an external force applied to the container, the discharge cap comprising: a nozzle that forms a nozzle flow path with a sectional area smaller than an opening area of the mouth part; and a valve that is provided on the nozzle and that is made of an elastic material softer than the nozzle, wherein the valve includes a valve part through which the content liquid in the nozzle flow path is discharged, a tubular held part that is held on the nozzle by a holder, and a tubular connecting part that connects the valve part and the held part to each other, the valve part includes a slit part provided with a slit opening and closing according to a difference between an atmospheric pressure and a pressure in the interior space of the container, and an outer circumferential part coupled to the connecting part, and the slit part more easily deforms with a pressure change in the interior space than the outer circumferential part, deforms in a discharge direction of the content liquid at a time of discharge of the content liquid, and deforms toward the nozzle flow path after the content liquid is discharged.
  2. The discharge cap according to claim 1, wherein the slit part is integrally molded of a same elastic material as that of the outer circumferential part, and is thinner than the outer circumferential part.
  3. The discharge cap according to claim 2, wherein the valve part includes an intermediate part between the slit part and the outer circumferential part, and the intermediate part is gradually thinned from the outer circumferential part to the slit part.
  4. The discharge cap according to claim 3, wherein the valve part has an outer surface being a spherical concave face depressed toward the nozzle flow path, and an inner surface being a flat face.
  5. The discharge cap according to claim 1, wherein the slit part has a thickness equal to or less than a half of a thickness of the outer circumferential part, and the slit part is located on a side nearer the nozzle flow path in relation to an intermediate position of the outer circumferential part in a thickness direction.
  6. The discharge cap according to claim 1, wherein the connecting part more easily deforms with a pressure change in the interior space than the outer circumferential part, extends in the discharge direction at a time of discharge of the content liquid, and contracts in a returning direction after the content liquid is discharged.
  7. The discharge cap according to claim 1, wherein the connecting part more easily deforms with a pressure change in the interior space than the outer circumferential part, and is inclined in a circumferentially inward direction of the connecting part after the content liquid is discharged.
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